

Лабораторная работа № 5. Конфигурирование VLAN

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Цель работы

Получить основные навыки по настройке VLAN на коммутаторах сети.

Задание

1. На коммутаторах сети настроить Trunk-порты на соответствующих интерфейсах, связывающих коммутаторы между собой.
2. Коммутатор msk-donskaya-sw-1 настроить как VTP-сервер и прописать на нём номера и названия VLAN.
3. Коммутаторы msk-donskaya-sw-2 — msk-donskaya-sw-4, msk-pavlovskaya-sw-1 настроить как VTP-клиенты, на интерфейсах указать принадлежность к соответствующему VLAN.

Задание

4. На серверах прописать IP-адреса.
5. На конечных устройствах указать соответствующий адрес шлюза и прописать статические IP-адреса из диапазона соответствующей сети, следуя регламенту выделения ip-адресов.
6. Проверить доступность устройств, принадлежащих одному VLAN, и недоступность устройств, принадлежащих разным VLAN.
7. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

Настройка Trunk-портов

```
msk-donskaya-omabakumova-sw-1>enable
Password:
msk-donskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-1(config)#interface g0/1
% Invalid input detected at '^' marker.

msk-donskaya-omabakumova-sw-1(config)#interface g0/1
msk-donskaya-omabakumova-sw-1(config-if)#switchport mode trunk

msk-donskaya-omabakumova-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-donskaya-omabakumova-sw-1(config-if)#exit
msk-donskaya-omabakumova-sw-1(config)#exit
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
```

Рис. 1: Настройка Trunk-порта для msk-donskaya-omabakumova-sw-1

Настройка Trunk-портов

```
msk-donskaya-omabakumova-sw-2>enable
Password:
msk-donskaya-omabakumova-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-2(config)#interface g0/1
msk-donskaya-omabakumova-sw-2(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-2(config-if)#interface g0/2
msk-donskaya-omabakumova-sw-2(config-if)#switchport mode trunk

msk-donskaya-omabakumova-sw-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-omabakumova-sw-2(config-if)#exit
msk-donskaya-omabakumova-sw-2(config)#exit
msk-donskaya-omabakumova-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-2#
```

Рис. 2: Настройка Trunk-порта для msk-donskaya-omabakumova-sw-2

Настройка Trunk-портов

```
msk-donskaya-omabakumova-sw-3>enable
Password:
msk-donskaya-omabakumova-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-3(config)#interface g0/1
^
% Invalid input detected at '^' marker.

msk-donskaya-omabakumova-sw-3(config)#interface g0/1
msk-donskaya-omabakumova-sw-3(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-3(config-if)#exit
msk-donskaya-omabakumova-sw-3(config)#exit
msk-donskaya-omabakumova-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-3#
```

Рис. 3: Настройка Trunk-порта для msk-donskaya-omabakumova-sw-3

Настройка Trunk-портов

```
User Access Verification
Password:
msk-donskaya-omabakumova-sw-4>enable
Password:
msk-donskaya-omabakumova-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-4(config)#interface g0/1
msk-donskaya-omabakumova-sw-4(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-4(config-if)#exit
msk-donskaya-omabakumova-sw-4(config)#exit
msk-donskaya-omabakumova-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-4#
```

Рис. 4: Настройка Trunk-порта для msk-donskaya-omabakumova-sw-4

Настройка Trunk-портов

```
msk-donskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-1#
msk-donskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-1(config)#interface g0/1
msk-donskaya-omabakumova-sw-1(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-1(config-if)#exit
msk-donskaya-omabakumova-sw-1(config)#interface f0/24
msk-donskaya-omabakumova-sw-1(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-1(config-if)#exit
msk-donskaya-omabakumova-sw-1(config)#exit
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
```

msk-donskaya-omabakumova-sw-1#

Рис. 5: Настройка Trunk-порта(дополнительно) для msk-donskaya-omabakumova-sw-1

Настройка Trunk-портов

```
-----
Password:
msk-pavlovskaya-omabakumova-sw-1>enable
Password:
msk-pavlovskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-omabakumova-sw-1(config)#interface fa0/24
msk-pavlovskaya-omabakumova-sw-1(config-if)#switchport mode trunk
msk-pavlovskaya-omabakumova-sw-1(config-if)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#exit
msk-pavlovskaya-omabakumova-sw-1#
%SYS-5-CONFIG-I: Configured from console by console

msk-pavlovskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-omabakumova-sw-1#
```

Рис. 6: Настройка Trunk-порта для msk-pavlovskaya-omabakumova-sw-1

```
msk-donskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-omabakumova-sw-1(config)#vtp domain dontskaya
Changing VTP domain name from NULL to dontskaya
msk-donskaya-omabakumova-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-omabakumova-sw-1(config)#vlan 2
msk-donskaya-omabakumova-sw-1(config-vlan)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-omabakumova-sw-1(config-vlan)#name management
msk-donskaya-omabakumova-sw-1(config-vlan)#vlan 3
msk-donskaya-omabakumova-sw-1(config-vlan)#name servers
msk-donskaya-omabakumova-sw-1(config-vlan)#vlan 101
msk-donskaya-omabakumova-sw-1(config-vlan)#name dk
msk-donskaya-omabakumova-sw-1(config-vlan)#vlan 102
msk-donskaya-omabakumova-sw-1(config-vlan)#name departments
msk-donskaya-omabakumova-sw-1(config-vlan)#vlan 103
msk-donskaya-omabakumova-sw-1(config-vlan)#name adm
msk-donskaya-omabakumova-sw-1(config-vlan)#vlan 104
msk-donskaya-omabakumova-sw-1(config-vlan)#name other
msk-donskaya-omabakumova-sw-1(config-vlan)#
```

Рис. 7: Настройка VLAN для msk-donskaya-omabakumova-sw-1

```
msk-donskaya-omabakumova-sw-1#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
2	management	active	
3	servers	active	
101	dk	active	
102	departments	active	
103	adm	active	
104	other	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgNo	Stp	BrdgMode	Trans1	Trans2
--More--										

Рис. 8: Вывод команды show vlan

```
msk-donskaya-omabakumova-sw-1#enable
msk-donskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-1(config)#vtp domain dontskaya
Domain name already set to dontskaya.
msk-donskaya-omabakumova-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-omabakumova-sw-1(config)#vtp password cisco
Password already set to cisco
msk-donskaya-omabakumova-sw-1(config)#exit
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-1#
```

Рис. 9: Настройка msk-donskaya-omabakumova-sw-1, как VTP-сервера

```
msk-donskaya-omabakumova-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-2(config)#vtp domain dontskaya
Domain name already set to dontskaya.
msk-donskaya-omabakumova-sw-2(config)#vtp mode client
Device mode already VTP CLIENT.
msk-donskaya-omabakumova-sw-2(config)#vtp password cisco
Password already set to cisco
msk-donskaya-omabakumova-sw-2(config)#exit
msk-donskaya-omabakumova-sw-2#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-omabakumova-sw-2#
```

Рис. 10: Настройка msk-donskaya-omabakumova-sw-2, как VTP-клиента

```
msk-donskaya-omabakumova-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-3(config)#vtp domain donskeya
Domain name already set to donskeya.
msk-donskaya-omabakumova-sw-3(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-omabakumova-sw-3(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-omabakumova-sw-3(config)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up
msk-donskaya-omabakumova-sw-3(config)#
```

Рис. 11: Настройка msk-donskaya-omabakumova-sw-3, как VTP-клиента

```
msk-donskaya-omabakumova-sw-4>enable
Password:
msk-donskaya-omabakumova-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-4(config)#vtp domain donskeya
Domain name already set to donskeya.
msk-donskaya-omabakumova-sw-4(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-omabakumova-sw-4(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-omabakumova-sw-4(config)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up
```

Рис. 12: Настройка msk-donskaya-omabakumova-sw-4, как VTP-клиента

```
msk-donskaya-omabakumova-sw-4#show vtp
% Incomplete command.
msk-donskaya-omabakumova-sw-4#show vtp status
VTP Version                : 1
Configuration Revision      : 12
Maximum VLANs supported locally : 255
Number of existing VLANs    : 11
VTP Operating Mode          : Client
VTP Domain Name             : donskaya
VTP Pruning Mode            : Disabled
VTP V2 Mode                 : Disabled
VTP Traps Generation        : Disabled
MDS digest                  : 0x2A 0xEA 0xA7 0xCC 0xAD 0xCC 0xB9 0x92
Configuration last modified by 10.128.1.2 at 3-1-93 02:54:07
msk-donskaya-omabakumova-sw-4#
```

Рис. 13: Вывод команды show vtp

```
msk-donskaya-omabakumova-sw-4#show vlan
VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/2
2    management             active
3    servers                active
101  dk                      active
102  departments             active
103  adm                     active
104  other                   active
1002 fddi-default           active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

VLAN Type  SAID      MTU    Parent RingNo BridgeNo Stp    BrdgMode Trans1 Trans2
-----
msk-donskaya-omabakumova-sw-4#
```

Рис. 14: Вывод команды show vlan на msk-donskaya-omabakumova-sw-4

```
msk-pavlovskaya-omabakumova-sw-1>enable
Password:
msk-pavlovskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-omabakumova-sw-1(config)#vtp domain donskaya
Domain name already set to donskaya.
msk-pavlovskaya-omabakumova-sw-1(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-pavlovskaya-omabakumova-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-pavlovskaya-omabakumova-sw-1(config)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up
```

Рис. 15: Настройка msk-pavlovskaya-omabakumova-sw-1, как VTP-клиента

Настройка конфигурации портов

```
msk-donskaya-omabakumova-sw-4>enable
Password:
Password:
msk-donskaya-omabakumova-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-4(config)#vtp mode client
Device mode already VTP CLIENT.
msk-donskaya-omabakumova-sw-4(config)#interface range f0/1 - 5
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport mode access
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport access vlan 101
msk-donskaya-omabakumova-sw-4(config-if-range)#exit
msk-donskaya-omabakumova-sw-4(config)#interface range f0/6 - 10
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport mode access
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport access vlan 102
msk-donskaya-omabakumova-sw-4(config-if-range)#exit
msk-donskaya-omabakumova-sw-4(config)#interface range f0/11 - 15
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport mode access
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport access vlan 103
msk-donskaya-omabakumova-sw-4(config-if-range)#exit
msk-donskaya-omabakumova-sw-4(config)#interface range f0/16 - 24
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport mode access
msk-donskaya-omabakumova-sw-4(config-if-range)#switchport access vlan 104
msk-donskaya-omabakumova-sw-4(config-if-range)#exit
msk-donskaya-omabakumova-sw-4(config)#exit
msk-donskaya-omabakumova-sw-4#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-omabakumova-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-4#
```

Рис. 16: Настройка конфигурации диапазона портов для msk-donskaya-omabakumova-sw-4

Настройка конфигурации портов

```
msk-donskaya-omabakumova-sw-2>enable
Password:
msk-donskaya-omabakumova-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-2(config)#vtp mode client
Device mode already VTP CLIENT.
msk-donskaya-omabakumova-sw-2(config)#interface range f0/1 - 2
msk-donskaya-omabakumova-sw-2(config-if-range)#switchport mode access
msk-donskaya-omabakumova-sw-2(config-if-range)#switchport access vlan 3
msk-donskaya-omabakumova-sw-2(config-if-range)#exit
msk-donskaya-omabakumova-sw-2(config)#exit
msk-donskaya-omabakumova-sw-2#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-omabakumova-sw-2#
```

Рис. 17: Настройка конфигурации диапазона портов для msk-donskaya-omabakumova-sw-2

Настройка конфигурации портов

```
msk-donskaya-omabakumova-sw-3>enable
Password:
msk-donskaya-omabakumova-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-3(config)#vtp mode client
Device mode already VTP CLIENT.
msk-donskaya-omabakumova-sw-3(config)#interface range f0/1 - 2
msk-donskaya-omabakumova-sw-3(config-if-range)#switchport mode access
msk-donskaya-omabakumova-sw-3(config-if-range)#
msk-donskaya-omabakumova-sw-3(config-if-range)#switchport access vlan 3
msk-donskaya-omabakumova-sw-3(config-if-range)#exit
msk-donskaya-omabakumova-sw-3(config)#exit
msk-donskaya-omabakumova-sw-3#
%SYS-5-CONFIG_I: Configured from console by console
```

Рис. 18: Настройка конфигурации диапазона портов для msk-donskaya-omabakumova-sw-3

Настройка конфигурации портов

```
msk-pavlovskaya-omabakumova-sw-1>enable
Password:
msk-pavlovskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-omabakumova-sw-1(config)#vtp mode client
Device mode already VTP CLIENT.
msk-pavlovskaya-omabakumova-sw-1(config)#interface range f0/1 - 15
msk-pavlovskaya-omabakumova-sw-1(config-if-range)#switchport mode access
msk-pavlovskaya-omabakumova-sw-1(config-if-range)#switchport access vlan 101
msk-pavlovskaya-omabakumova-sw-1(config-if-range)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#interface range f0/20
msk-pavlovskaya-omabakumova-sw-1(config-if-range)#switchport mode access
msk-pavlovskaya-omabakumova-sw-1(config-if-range)#switchport access vlan 104
msk-pavlovskaya-omabakumova-sw-1(config-if-range)#exit
msk-pavlovskaya-omabakumova-sw-1(config)#exit
msk-pavlovskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Рис. 19: Настройка конфигурации диапазона портов для msk-pavlovskaya-omabakumova-sw-1

Настройка шлюзов и IP-адреса у оконечных устройств

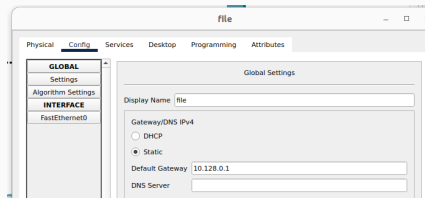


Рис. 20: Настройка шлюза для оконечного устройства

Настройка шлюзов и IP-адреса у оконечных устройств

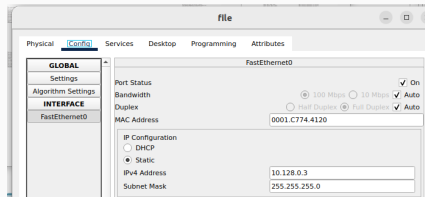
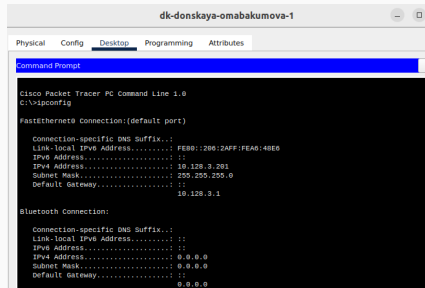


Рис. 21: Настройка IP-адреса для оконечного устройства

Пингование адресов



The screenshot shows a Cisco Packet Tracer PC Command Line 1.0 window. The window title is "dk-donskaya-omabakumova-1". The "Desktop" tab is selected. The command prompt shows the output of the "ipconfig" command. The output displays network configuration for the FastEthernet0 interface, including the Link-local IPv6 Address (FE80::206:2AFF:FEA0:48E0), IPv6 Address (10.128.3.201), Subnet Mask (255.255.255.0), and Default Gateway (10.128.3.1). It also shows the Bluetooth Connection configuration, which is currently empty.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::206:2AFF:FEA0:48E0
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.128.3.201
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   10.128.3.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

Рис. 22: Вывод команды ipconfig

Пингование адресов

```
c:\>ping 10.128.3.202

Pinging 10.128.3.202 with 32 bytes of data:

Reply from 10.128.3.202: bytes=32 time<1ms TTL=128
Reply from 10.128.3.202: bytes=32 time<1ms TTL=128
Reply from 10.128.3.202: bytes=32 time<1ms TTL=128
Reply from 10.128.3.202: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.202:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

c:\>ping 10.128.4.202

Pinging 10.128.4.202 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.4.202:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

c:\>
```

Рис. 23: Пингование IP-адресов

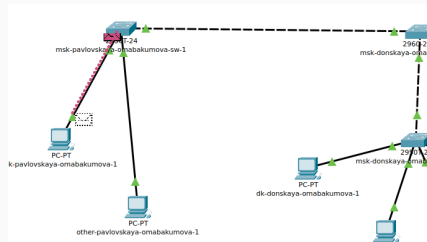


Рис. 24: Пакеты успешно проходят

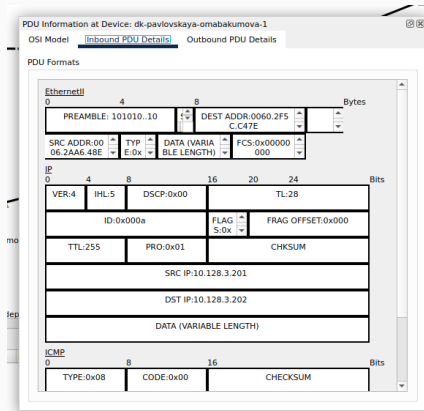


Рис. 25: Информация PDU

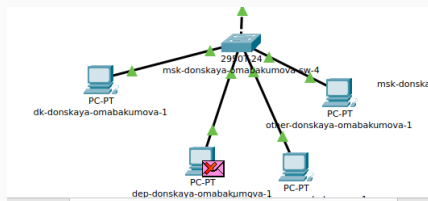


Рис. 26: Пакеты не проходят

Выводы

Во время выполнения данной лабораторной работы я приобрела основные навыки по настройке VLAN на коммутаторах сети.